

QUANTITATIVE DESCRIPTION OF ENVIRONMENT-BEHAVIOR RELATIONS

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In 1687, Sir Isaac Newton invented a profound new way of thinking about how objects move across space and time. Newton tackled familiar empirical facts such as the observation that objects fall when dropped, but the description of falling objects was elevated to a new realm of precise, quantitative analysis. Among other feats, his analysis produced the universal law of gravitation, which yielded considerable predictive and practical advantages over previous accounts (Kline, 1959). The stunning successes of classical mechanics, of landing rockets on the moon, would not have been possible before Newton's framework. In addition to its predictive and practical advantages, the universal law of gravitation unified seemingly disparate phenomena: It described moving bodies both here on Earth and in the heavens. Newton's account also provided a foundation for attempts to explain gravity. As sciences have matured, from astronomy to zoology, so too has their use of quantitative analysis and their ability to describe, unify, and explain. The field of behavior analysis has witnessed similar transformations in how people describe an organism's behavior across space and time.

Fortunately, one does not need Newton's calculus to understand and appreciate these exciting advances in behavioral science. In this chapter, we explain key techniques involved in quantitative analysis. We describe how quantitative models are evaluated and compared. To provide some theoretical backbone to our explication of techniques, we use a model of choice known as matching theory as

a case study in quantitative analysis. Although we have selected matching theory as our case study because of its widespread application in behavior analysis, the techniques and issues we discuss could be generalized to any quantitative model of environment-behavior relations, and where appropriate we highlight these extensions.

ON THE UTILITY OF QUANTITATIVE MODELS

Models specify relations among dependent variables and one or more independent variables. These relations can be specified using words (a verbal model) or mathematics (a mathematical or quantitative model). Verbal models can be just as useful as quantitative models in describing the causes of behavior. For instance, an applied behavior analyst does not need quantitative models to assess determinants of self-injurious behavior or to generate verbal behavior for a child with autism. However, there are also examples in which quantitative models are useful in the applied realm (Critchfield & Reed, 2009; Fisher & Mazur 1997; Mazur, 2006; McDowell, 1982). Quantitative models of choice can help the analyst tease apart controlling variables in a naturalistic setting (McDowell, 1982; see Fuqua, 1984, for some caveats to this assertion), or they may be useful in terms of evaluating preferences for treatment options (Fisher & Mazur, 1997). Even if the benefits are not immediate, knowledge of quantitative accounts could lead to alternative treatments

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(McDowell, 1982), or it could inspire new insights into the determinants of problem behavior (Critchfield & Kollins, 2001; Fisher & Mazur, 1997; Nevin & Grace, 2000; see Volume 2, Chapters 5 and 7, this handbook).

Behavioral scientists should appreciate and understand a wide variety of analytic methods to understand the determinants of behavior. Quantitative analysis entails a rich and powerful set of tools. From 1970 to 2000, basic behavior-analytic science saw an increase from 10% to almost 30% in articles published in the *Journal of the Experimental Analysis of Behavior* that used equations to describe behavior (Mazur, 2006). In addition to understanding the advances described in these articles, other intellectual and practical payoffs are derived from knowledge of quantitative analysis. These benefits may be delayed, but they can be profound. The English philosopher Roger Bacon (c. 1214–1294, as quoted in Kline, 1959) noted,

Mathematics is the gate and the key of the sciences. . . . Neglect of mathematics works injury to all knowledge, since he who is ignorant of it cannot know the other sciences or the things of this world. And what is worse, men who are thus ignorant are unable to perceive their own ignorance and so do not seek a remedy. (p. 1)

Quantitative models have at least four benefits. First, quantitative models force a greater degree of precision than their verbal counterparts. Assume several reinforcer rates have been experimentally arranged, from low to high, and the response rates obtained at each reinforcer rate have been measured. A verbal description might assert that at low reinforcer rates, increases in reinforcer rate produce rapid increases in response rate, but that response rate eventually reaches a maximum at higher reinforcer rates. The same description of the relation between response rate and reinforcer rate is embodied by Herrnstein's hyperbolic equation (described in more detail later). Herrnstein's equation states succinctly and precisely how behavior will change with increases in reinforcement. Moreover, Herrnstein's equation precisely asserts how reinforcement from the experimentally arranged schedule of

reinforcement interacts with other, background sources of reinforcement. Assumptions about and descriptions of interactions between variables can be particularly troublesome and obscure if formulated in verbal terms. Quantitative models force specificity about the nature of these interactions. For example, Staddon (1984) analyzed a model of social dynamics by expressing the model quantitatively. Staddon's quantitative analysis revealed several inconsistencies (and even contradictions) in the corresponding verbal formulation. Staddon concluded that "unaided verbal reasoning is almost useless in the analysis of the dynamics of interactions, human or mechanical" (p. 507).

Second, the predictions of a quantitative model are more specific than the predictions of a verbal description, which allows one to observe minor but systematic deviations from the predictions. For instance, one could verbally describe where a projectile might land after being launched into the air (e.g., "over there"), or one could use an equation to make a precise prediction (e.g., 157 feet, 2 inches, due north). Any deviation from the latter location may be precisely measured, and records may be kept to determine whether the equation makes systematic errors (e.g., it consistently overestimates the distance traveled). Similarly, one could hypothesize verbally about how much reinforcement would be necessary to establish some level of appropriate behavior (e.g., one might say one needs to identify a "powerful" reinforcer or deliver the reinforcer "frequently"). As with the estimate of the projectile, a hypothesis might be qualitatively correct, but it will be more precise if one uses established equations of reinforced responding. Admittedly, this degree of precision may not be necessary in an applied context. Nevertheless, a hallmark of behavioral science is the predictive power of its explanations, and, as these examples imply, quantitative models surpass verbal models in making predictions.

Third, to the extent that the predictions and conditions under which one should observe them are more precise, the more falsifiable the theory becomes. As the philosopher of science Karl Popper (1962) noted, falsifiability is a virtue of scientific theory. Consider Sir Arthur Eddington's test of Albert Einstein's theory of general relativity. One of

general relativity theory's critical predictions is that gravity bends light. To test this prediction, Eddington measured the amount of shift in light from stars close to the sun (Silver, 1998). The sun's powerful gravitational field, according to general relativity theory, should produce measurable shifts in the light emanating from the nearby stars. Eddington waited for an eclipse (the eclipse of 1919), when the stars adjacent to the sun were observable, and he measured the amount of shift in light emanating from those stars. He found not only that the light from these stars did indeed bend around the sun, but also that the exact amount of shift was as predicted by Einstein's theory. An observation (or more realistically, a series of observations) to the contrary would have falsified the theory. Similarly, Herrnstein (1974) noted that certain predictions made by his quantitative theory of reinforced responding were especially vulnerable to "empirical confirmation or refutation" (p. 164). All good scientific theories, whether verbal or quantitative, are falsifiable. One virtue of most quantitative theories is that the conditions under which one can refute them are relatively clear (McDowell, 1986).

Fourth, quantitative modeling encourages researchers to unify diverse phenomena; to see similarities in the determinants of behavior across settings, cultures, and species (Critchfield & Reed, 2009; Lattal, 2001; Mazur, 2006; see Chapter 7, this volume). Another way to put this is that a goal of a science of behavior is to discover invariance, or regularity in nature. For example, people see the diversity of structure in the animal kingdom as a result of evolutionary processes and the diversity of geological events as a result of tectonic processes. The mathematician Bell (1945, p. 420) defined *invariance* as "changelessness in the midst of change, permanence in a world of flux, the persistence of configurations that remain the same despite the swirl and stress of countless transformations" (also quoted in Nevin, 1984, p. 422). Skinner (1950/1972) also saw the discovery of invariance as a worthy goal:

Beyond the collection of uniform relationships lies the need for a formal

representation¹ of the data reduced to a minimal number of terms. A theoretical construction may yield greater generality than any assemblage of facts. It will not stand in the way of our search for functional relations because it will arise only after relevant variables have been found and studied. Though it may be difficult to understand, it will not be easily misunderstood. (p. 100).

Skinner's (1950/1972) assessment of theory, however, was tempered by a recommendation that psychologists must first establish an experimental analysis of how relevant variables affect behavior. Ultimately, however, quantitative theory increases the precision and generality of explanations and improves the ability to predict and influence behavior.

STRUCTURE AND FUNCTION OF QUANTITATIVE MODELS

Although quantitative models can generate new predictions and explanations of behavior, they are often developed inductively on the basis of descriptions of empirical facts, and this is where we start. After rigorous, parametric, experimental analysis, researchers plot a dependent variable (e.g., response rate, interresponse times, ratio of time spent engaged in one activity over another activity) as a function of an independent variable (e.g., reinforcer rate, time in session, reinforcers delivered for one activity relative to another activity) in graphical space. They examine the shape of the relation. Is it described by a straight line or a curve? In behavioral science, relations characterized by straight lines or monotonic curves (curves that change in one direction) are common. Researchers may also model behavioral processes that show a bitonic relation, which is a curve that changes in two directions (e.g., a curve that increases and then decreases or vice versa). Although these shapes are not complicated, the equations that describe them may appear to be. If the equations are intimidating at first, start with the shapes and the specific environment–behavior

¹A formal representation means a mathematical representation.

relations described by the shapes. Therefore, use careful visual inspection (see Chapter 9, this volume) of the data.

The importance of visual analysis is illustrated nicely by Anscombe's quartet (Anscombe, 1973), which is a series of four datasets that are shown in Figure 10.1. Across the four panels of the figure, the mean and variance of the data points in each panel are equivalent, as is the linear equation describing the relation between the independent (x-axis) and dependent (y-axis) variables: $y = 0.5x + 3$. Even a casual visual analysis, however, reveals that the shape of each dataset is remarkably distinct and that only the data in the upper left panel are appropriately described by a linear equation. A purely quantitative analysis alone is not sufficient. Careful use of quantitative analysis involves careful visual inspection of the data (Parsonson & Baer, 1992; see Chapter 9, this volume), not to mention rigorous experimental analysis (Perone, 1999; Sidman, 1960; see Chapter 5, this volume).

After a visual analysis of the relation between independent and dependent variables, one must

dissect the anatomy of the equation that describes the relation. In behavioral and psychology journals, equations may appear in the introduction of an article. Thus, dissecting the equation may require some detective work and skipping ahead in the article to find a graph that shows the shape described by the equation. Comparing the equation with the graph is useful. Here is an example of a common equation in behavioral science:

$$R = k \left(\frac{r}{r + r_c} \right). \quad (1)$$

This equation, known as *Herrnstein's hyperbolic equation*, describes a hyperbolic relation between reinforcer rate, r , and response rate, R . Figure 10.2 shows two examples of this hyperbolic shape described by the equation. The first step to understanding a new equation is to identify the variables in the equation, or the environmental and behavioral events that are measured directly in the experiment. In the case of Herrnstein's hyperbolic equation, the experimenter measures response rate,

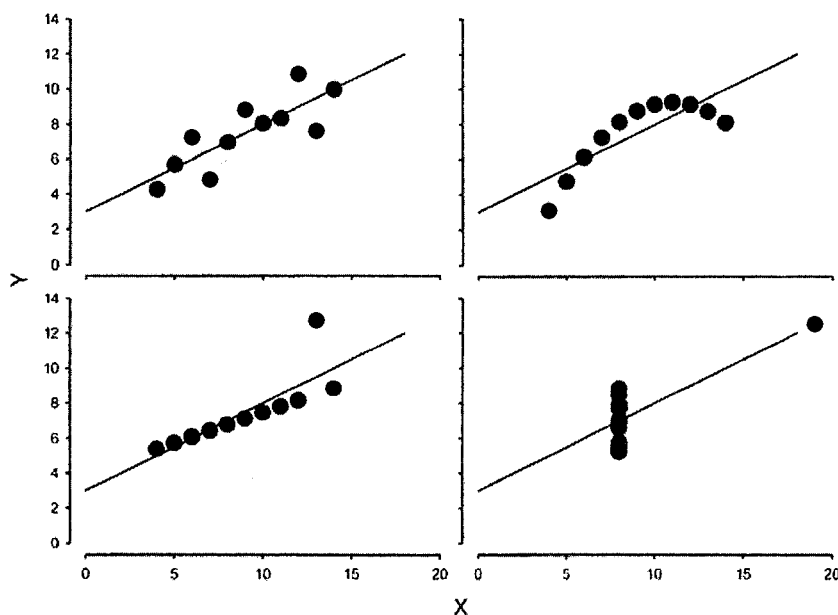


FIGURE 10.1. Anscombe's (1973) quartet. The four graphs show different data sets that are described by the same linear equation, which is indicated by the straight line in each panel. The datasets also have the same mean and variance. From "Graphs in Statistical Analysis," by F. J. Anscombe, 1973, *American Statistician*, 27, pp. 19–20. Reprinted with permission from *The American Statistician*. Copyright 1973 by the American Statistical Association. All rights reserved.

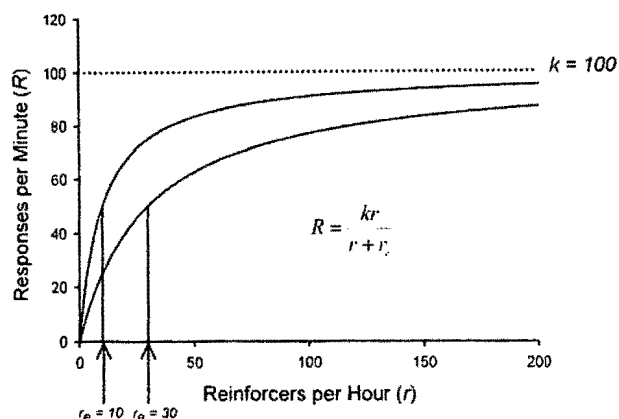


FIGURE 10.2. Two examples of Herrnstein's hyperbolic equation, which predicts how response rate changes with increases in reinforcer rate. The only difference between the two curves is the value of r_e .

R , as a function of reinforcer rate, r . For example, assume one measured rates of engaging in disruptive behavior as a function of rates of social attention for this behavior. The dependent variables (e.g., response rates) will always appear on the left side of the equation, and the independent variables (e.g., reinforcer rates) will always appear on the right side of the equation.

Next, identify the parameters. The parameters are the quantities in the equation that are not directly measured; rather, they are estimated statistically (see Evaluating Linear Models: Linear Regression and Evaluating Linear Models: Nonlinear Regression sections). The parameters affect the shape and position of the curve (or line), and a computer program that conducts regression analyses will find values for the parameters so that the curve (or line) comes as close to the data as possible (called best-fit parameter values). To illustrate how the value of a parameter affects the shape of the curve in Figure 10.2, we used two values for the parameter r_e and the same value for the parameter k . If one considers just the top curve, for example, the curve predicts response rates given one value of k , one value of r_e , and all reinforcer rates between zero and 200 reinforcers per hour. The curve can also predict response rates for higher reinforcer rates, but for practical purposes, we do not show these in the graph. The value of r_e is small in the top curve. In terms of the equation, the value of r_e does not produce a large impact in the denominator, and so

response rates increase rapidly with increases in reinforcer rates. As r_e increases, however, as in the bottom curve, the effects of increases in reinforcer rates are essentially dampened. Because r and r_e are summed in the denominator, smaller changes occur in R with increases in reinforcer rates.

Parameters usually reflect the operation of environmental, behavioral, or physiological processes. For example, the parameter r_e in Herrnstein's hyperbolic equation is thought to reflect the rate of background, extraneous reinforcers. In our example, these extraneous reinforcers represent all of the reinforcers in the child's environment (video games, food, etc.) except for the reinforcers delivered as a consequence of disruptive behavior. The degree to which the interpretations of parameters are confirmed empirically is an interesting and exciting feature of quantitative modeling. As we discuss in more detail in the Evaluating Nonlinear Models: Nonlinear Regression section, even if an equation describes a dataset extremely well (e.g., the curve comes very close to the obtained data), it does not necessarily mean that the interpretations of the parameters are supported. Eventually, such inferences should be supported by direct experimental analysis of the controlling variables, whether biological or environmental (Dallery & Soto, 2004; Shull, 1991). For instance, if r_e is thought to measure extraneous reinforcers, its value should increase or decrease as reinforcers are added or subtracted, respectively, from the environment in which the target behavior is measured.

One way to understand an equation is to plot the equation using multiple parameter values as in Figure 10.2. Creating multiple plots of an equation can easily be done using a spreadsheet program such as Microsoft Excel (at the time of this writing, Excel is perhaps the most common spreadsheet program in use). As described earlier, Figure 10.2 illustrates that the smaller the value of r_e , the steeper the curve rises toward its asymptote, k . Although not shown, plotting Equation 1 with different values of k illustrates how increases in k increase the maximum value reached by the curve (100 in Figure 10.2). Understanding how the shape of the function changes as the parameter values change is essential because changes in the shape reflect changes in

behavior. In an applied context, increasing r_c as depicted in Figure 10.2 would mean that rates of disruptive behavior would decrease, even if reinforcer rates for disruptive behavior remain the same (compare the predicted response rates between curves when the reinforcer rate is 100 reinforcers/hour; McDowell, 1986).

There is obviously more to dissecting and digesting a quantitative model than examining the shape defined by the equation. Researchers need to ask whether they have enough data to evaluate the model (e.g., a general rule of thumb is to have twice as many data points as the number of parameters), whether there are known properties of behavior or alternative models that they need to consider in evaluating the model, and whether they need to consider statistical or theoretical requirements for estimated parameter values (we discuss these considerations in more detail in the Evaluating Linear Models: Linear Regression and Evaluating Nonlinear Models: Nonlinear Regression sections).

This brief introduction provides a starting point from which to approach equations in behavioral science. All equations in behavioral science share the same general structural and functional characteristics. The tactics presented in this section represent a broad, yet effective strategy to dissect equations. They may be summarized as follows: Visually inspect the shape of the dataset when plotted in graphical space, decompose the variables and parameters in the equation, explore how the plot of the equation changes as the parameter values change, and consider what these changes mean in terms of behavior and its causes.

INTRODUCTION TO A QUANTITATIVE DESCRIPTION OF CHOICE

One purpose of quantitative analysis is to understand the complex interaction between the determinants of behavior (e.g., biological, environmental, pharmacological) and some measure of behavior (e.g., response rate, latency). For instance, researchers may be interested in when and why a bee travels to a new field of flowers for nectar, why a shopper on a diet picks up some candy in the checkout line, or why a pigeon in an operant chamber impulsively

pecks a lighted key for an immediately delivered small amount of grain instead of waiting for a larger amount. These examples of choice, of choosing one activity over another, are amenable to quantitative analysis. One common way in which psychologists study choice in the operant laboratory is by using concurrent schedules of reinforcement. In the pigeon example, two concurrent, mutually exclusive alternatives were available (pecking for small payoffs or waiting for larger payoffs). This situation is also common in naturalistic, clinical settings: A child may choose to engage in disruptive behavior rather than appropriate behavior (Borrero & Vollmer, 2002); a smoker may choose to light up or forgo a cigarette (Dallery & Raiff, 2007).

The relative rates at which the behavior occurs—in the laboratory or in the world outside the laboratory—can be powerfully governed by relative rates of reinforcement earned for engaging in them. To quantitatively model the relation between relative responding and reinforcement over time, the response and reinforcer rates need to be denoted in mathematical terms. In the case of the pigeon responding on a two-alternative concurrent schedule of reinforcement, the rate of responding on one alternative can be represented by R_1 , and the rate of responding on the other by R_2 . The rate at which reinforcers are delivered on the two alternatives can be represented by r_1 and r_2 . Some authors use different letters, but they signify the same quantities.

In 1961, Richard Herrnstein examined the relation between rates of reinforcement and responding in a seminal study. His subjects were pigeons, and they could peck two lighted keys that were mounted on the front panel (one on the left and one on the right) of an experimental operant chamber. Pecking the keys sometimes resulted in brief access to food. Specifically, the rates of food access varied according to variable-interval (VI) schedules of reinforcement, which delivered a reinforcer for the first response after some interval had elapsed. For example, the left key may have been scheduled to provide access to food every 30 seconds on average, and the right key may have only provided access to food once every 60 seconds on average. After varying these rates of reinforcement across a wide range of values, Herrnstein plotted the proportion of responses allocated to one

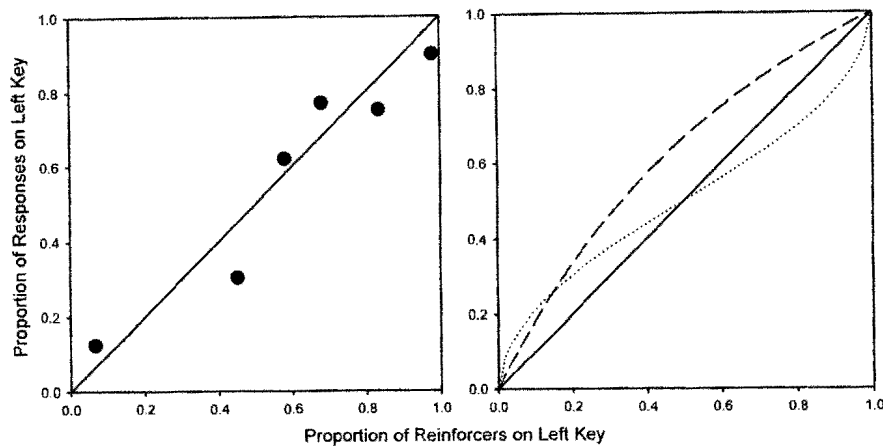


FIGURE 10.3. Examples of matching and deviations from matching. The left panel shows perfect matching. The right panel shows two common deviations from perfect matching. The dashed curve is an example of bias, and the dotted curve represents an example of insensitivity.

key (in our example, responding on the left key is designated as R_1) as a function of the proportion of reinforcers delivered on that key. Herrnstein found that the shape of the data path was a straight line. See the diagonal line in the left panel of Figure 10.3. Data points indicate hypothetical obtained response and reinforcer proportions. The straight line predicts that the proportion of responses should equal, or match, the proportion of reinforcers obtained at that alternative. So, for example, if the proportion of reinforcers on the left key equaled .25, then the proportion of behavior allocated to that side would also equal .25. This shape can be described by an equation:

$$\frac{R_1}{R_1 + R_2} = \frac{r_1}{r_1 + r_2} \quad (2)$$

Equation 2 describes a straight line passing through the origin ($x = 0, y = 0$) with a slope of 1—a relation termed *perfect matching*. Herrnstein's (1961) study, and Equation 2, has inspired hundreds of studies on choice. The applications have ranged from the laboratory to clinical settings (Borrero & Vollmer, 2002; Murray & Kollins, 2000; St. Peter et al., 2005; Taylor, Lincoln, & Foster, 2010; see Volume 2, Chapter 7, this handbook), social dynamics (Borrero et al., 2007; Conger & Killeen, 1974), and education (Martens & Houk, 1989). The matching law has also served as a basis for studies in behavioral ecology (Houston, 1986) and behavioral neuroscience (Lau & Glimcher, 2005).

As the predictions of Equation 2 were compared with behavior, however, researchers soon discovered that perfect matching was actually a special case of a more generalized equation (Baum, 1974; Staddon, 1968). Obtained data tended to deviate in two ways from the straight line shown in the left panel of Figure 10.3. First, animals sometimes showed a bias for one alternative over another. Perhaps one response key was easier to peck or, in a naturalistic setting, the attention derived from disruptive behavior was “better” than the minor attention for appropriate behavior. In the right panel of Figure 10.3, the impact of such asymmetries on behavior is shown by the dashed curve. Because the curve is bent upward, one can deduce in the context of our example that left responses are occurring more frequently than predicted by Equation 2 (the solid line). Note that this bias toward the left alternative is not caused by the rates of reinforcement associated with the left and right alternatives. Second, animals also showed some degree of insensitivity to changes in the rates of reinforcement. The dotted curve shows an example of such insensitivity; note how the curve bends toward indifference, or a proportion of .5. The curve is above the straight line of equality at x-axis values less than .5 and below the straight line at x-axis values above .5.

A single equation describing sensitivity to relative reinforcer rate across the two alternatives and bias toward one alternative over the other (not

resulting from differences in reinforcer rate) would be useful. Unfortunately, bias and sensitivity are hard to detect visually and to quantify when proportions are used. They are hard to detect because a plot of an animal's behavior may show biased responding (e.g., the plot tends to bend upward) and insensitivity (the plot also bends toward indifference) at the same time. Also, data from live organisms usually demonstrate more variability than our graphical example, which further complicates visual detection of bias and insensitivity. Staddon (1968) found that plotting ratios instead of proportions aided in visually detecting deviations from strict matching. The ratio form of matching is algebraically equivalent to the proportional form (to obtain the ratio form from the proportional form, invert both sides of Equation 2, separate the terms involved, and subtract one from both sides), and it is written

$$\frac{R_1}{R_2} = \frac{r_1}{r_2} \quad (3)$$

Baum (1974) modified Equation 3 to account for deviations from matching by adding two parameters, a and b , to Equation 3:

$$\frac{R_1}{R_2} = b \left(\frac{r_1}{r_2} \right)^a \quad (4)$$

where a quantifies sensitivity to the ratio of reinforcer rates on alternatives 1 and 2, and b quantifies bias. Equation 4 is known as the generalized matching equation, or power-function matching, because the independent variable is raised to a power, a . Traditionally, Equation 4 is evaluated by plotting the logarithm of the response ratio as a function of the logarithm of the reinforcer ratio. Logarithmic transformation² of Equation 4 results in a straight line when the equation's predictions are plotted in graphical space. The logarithmic form of the generalized matching equation is

$$\log \left(\frac{R_1}{R_2} \right) = a \log \left(\frac{r_1}{r_2} \right) + \log b, \quad (5)$$

where a is the slope, $\log b$ is the intercept, and the logged response ratio and logged reinforcer ratios are the dependent and independent variables, respectively. This straight line is different from the line described by Equation 2 in that its slope and intercept can vary; perfect matching specifies a slope of 1 and an intercept of 0.

Because the logarithmic form of the generalized matching equation is a straight line, one can evaluate the equation's descriptive accuracy by fitting a line to the data. Bias is indicated by the intercept of the straight line, $\log b$, and sensitivity is indicated by the slope of the line, a . One example of a line described by Equation 5 is shown in Figure 10.4. First, note that the slope is less than 1.0. This means that for every unit change in the reinforcer ratio, there is less than a one-unit change in the response ratio. Considering what the value of the slope means in terms of unit changes in the dependent and independent variables is often useful. Second, note that the line is elevated above the origin, which shows some bias for the response alternative in the numerator, which is R_1 , or the left key, in our example. Of course, the line could be steeper than 1.0, and it could intersect the ordinate below the origin. Consider what this

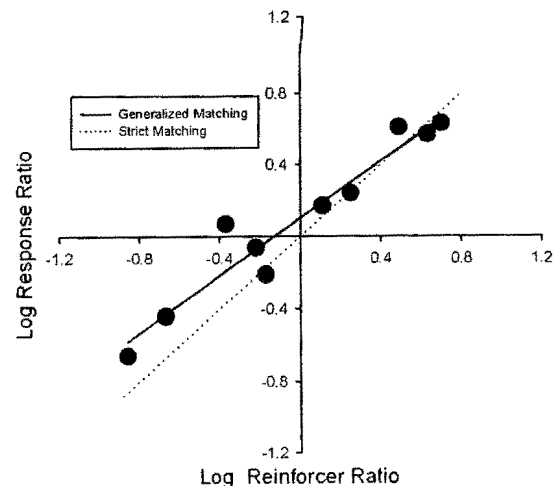


FIGURE 10.4. An example of the generalized matching equation. The line represents a best-fitting line obtained via least-squares regression to the obtained response and reinforcer rates. See the text for the equation. The data points in columns E and F from Table 10.2 are plotted.

²The logarithm of the product of two numbers is the sum of the logarithms of the individual numbers, and the logarithm of an exponentiated number is the product of the exponent and the logarithm of the number.

would mean in terms of behavior. It would mean that the change in response ratios is more than one unit for every unit change in reinforcer ratios (called *overmatching*; Baum, 1979) and that there is a bias toward the response alternative in the denominator, the right key. Now one has a single, elegant equation to describe how two variables (i.e., reinforcer rates on two alternatives), other asymmetries in the properties of the responses or reinforcers (i.e., changes in the intercept or bias), and the degree of insensitivity to changes in reinforcer rates (i.e., change in the slope or sensitivity) affect choice.

One log transforms the ratios to evaluate the generalized matching equation for several reasons. The main reason, as implied earlier, is that it is easier to detect departures from strict matching. The log transform also turns the curves produced by bias and insensitivity into straight lines (McDowell, 1989), and thus bias and insensitivity are reflected by changes in intercept and slope, respectively. Indeed, quantitative analysis has several useful data transformations, which are used for a variety of reasons, for example, to increase interpretability of a graph or to meet assumptions for statistical analysis (e.g., normality). Some balk at the idea of transforming behavioral data; after all, one does not observe logarithms of behavior. This argument misses the point of transforms, which is to magnify the ability to describe and explain environment–behavior relations. For instance, a logarithmic transform might help one to see the spread of data across the range of values of the independent variable, especially when that range is very large. That is, when plotting logarithms of the data, the spacing of data between numbers 1 and 10 is the same as the spacing between the numbers 10 and 100. Indeed, even a simple measure of response rate is a transformation. Transformations can have advantages, much as the metric system has advantages over the U.S. system of measurements (see Shull, 1991, and Tukey, 1977, for further discussion of transformations).

EVALUATING LINEAR MODELS: LINEAR REGRESSION

The generalized matching equation states that the relation between log reinforcer and log response

ratios can be described by a straight line—but is it? If visual inspection suggests that the data are orderly, sensible, and linear, one can evaluate this theory of choice statistically by performing linear regression. More generally, when researchers want to evaluate a relation between an independent and a dependent variable, and the relation between them can be graphed as a straight line, they use linear regression (if the variables are not linearly related, they use nonlinear regression, which is covered in the Evaluating Nonlinear Models: Nonlinear Regression section). When the two variables are not causally related, researchers use a correlation coefficient. For example, one would not want to perform regression to describe the relation between response rate and response latency. As a rule of thumb when deciding between regression and correlation, ask whether the independent variable plotted along the abscissa is causing the changes in the dependent variable. If it is, use regression. If the variable plotted along the abscissa is a dependent variable (response latency), and so is the variable plotted along the ordinate (e.g., response rate), then a correlation is more appropriate. Also, before conducting a so-called ordinary regression analysis, whether linear or nonlinear, it is important to ensure that the assumptions of ordinary regression are not violated. For convenience, these assumptions are listed in Table 10.1.

Consider the data shown in Table 10.2. These data show how a human might interact during a discussion with two confederates. The dependent variable is how much the individual interacts with each confederate or, more specifically, the rate of words per minute directed toward each confederate (columns C and D). The independent variable is the rate of social reinforcers per hour (e.g., agreement statements, nods; columns A and B) delivered by each of the confederates. The social reinforcers are delivered according to VI schedules. The rates of reinforcement are varied across 10 conditions. Thus, when the reinforcer rate is 5.97 reinforcers per hour for Confederate 1 and 4.63 reinforcers per hour for Confederate 2, the rates of words per minute directed toward each confederate are 6.84 and 4.66, respectively. It is important to remember the units in which the variables are expressed; they are important in understanding any quantitative model.

TABLE 10.1

Assumptions of Ordinary Least-Squares Regression

- The values of the independent variable are known precisely. All of the variability in the data is in the values of the dependent variable.
- The variability in the dependent variable, the error, follows a normal bell-shaped distribution.
- The degree of error, or scatter, in the dependent variable is the same at all points along the line (or curve). Another way to say this is that the standard deviation of the scatter must be the same at all points along the line (or curve), which is known as *homoscedasticity*.
- The observations are independent.

Note. From *Fitting Models to Biological Data Using Linear and Non-Linear Regression: A Practical Guide to Curve Fitting* (p. 57), by H. Motulsky and A. Christopoulos, 2004, Oxford, England: Oxford University Press. Copyright 2004 by Oxford University Press. Adapted by permission of Oxford University Press, Inc.

TABLE 10.2

Regression Calculations for a Linear Model

Condition	A: rnf rate by Conf. 1	B: rnf rate by Conf. 2	C: rsp rate to Conf. 1	D: rsp rate to Conf. 2	E: log rnf ratio	F: Log rsp ratio (Y)	G: predicted log rsp ratio (Y)	H: $Y - Y'$	I: $(Y - Y')^2$
1	5.97	4.63	6.84	4.66	.11	.17	.19	-0.02	0.0004
2	2.35	5.46	4.40	3.77	-.37	.07	-.19	0.26	0.0676
3	7.09	2.31	6.29	1.55	.49	.61	.49	0.12	0.0144
4	1.55	11.15	2.50	11.38	-.86	-.66	-.58	-0.08	0.0064
5	11.08	2.20	8.60	2.01	.70	.63	.66	-0.03	0.0009
6	2.42	3.55	1.20	1.96	-.17	-.21	-.03	-0.18	0.0324
7	4.96	1.17	2.06	0.55	.63	.57	.60	-0.03	0.0009
8	2.46	4.04	1.80	2.07	-.22	-.06	-.07	0.01	0.0001
9	4.57	2.55	2.48	1.42	.25	.24	.30	-0.06	0.0036
10	1.13	5.29	0.91	2.48	-.67	-.44	-.43	-0.01	0.0001

Note. Columns A–D show response and obtained reinforcer rates for a subject engaging in speech directed at Conf. 1 or 2. Columns E and F present the logged rsp. and rnf. rates, which are the data used to perform linear regression. Column G shows the predicted log rsp. ratios calculated by Equation 5 with $a = .80$ and $\log b = .10$. Column H shows the residuals, and column I shows the squared residuals. The sum of the squared residuals is $SS_{REG} = 0.13$, the sum of squared deviations from the mean is $SS_{TOT} = 1.78$. The r^2 is calculated as $1 - (SS_{REG}/SS_{TOT}) = .93$. Conf = confederate; rnf = reinforcement; rsp = response.

Columns E and F show the logged reinforcer ratios (r_1/r_2) and response ratios (R_1/R_2), respectively. The data from columns E and F are the data plotted graphically in Figure 10.4. The independent variable, the logged reinforcer ratio, is always plotted along the abscissa; it is important that the independent variable be plotted along the abscissa and the dependent variable be plotted on the ordinate when performing linear regression.

Linear regression may be calculated with a statistics program or by an algorithm in a spreadsheet program. These programs start with initial values of the slope and intercept and then adjust these values over a series of iterations so that the line is as close to the obtained data as possible.³ The initial values used by programs may be unknown to the user. If one uses a spreadsheet with regression capabilities, for example, one needs to enter the initial values

³The parameters of a line can also be calculated directly from formulas derived from linear regression theory rather than minimizing the residual sum-of-squares. We chose to illustrate the curve-fitting approach to calculating the parameter values because the logic should be easier to understand in the context of a more familiar function form (i.e., a line) relative to less familiar forms (i.e., curves) and because this method can be applied to both linear and nonlinear regression.

into the spreadsheet. Currently, for pedagogical purposes, assume the starting values must be entered. The starting values will depend on the particular dataset being evaluated. One shortcut in determining starting values is to inspect the plotted data. If the slope of the line is positive, one can use 1 as its starting value. If it is negative, one can use -1 . One can also examine where the line will intersect the ordinate at zero on the abscissa, and then enter that value as the starting value for the intercept. Once these values are entered, the equation makes predictions about the response ratios. For example, let us use an intercept of 0 and a slope of 1 and predict the logged response ratio when the logged reinforcer ratio is 0.11. Insert these values into the right side of Equation 5:

$$\log\left(\frac{R_1}{R_2}\right) = a \log\left(\frac{r_1}{r_2}\right) + \log b,$$

$$\log\left(\frac{R_1}{R_2}\right) = 1(0.11) + 0, \text{ and}$$

$$\log\left(\frac{R_1}{R_2}\right) = 0.11.$$

The predicted log response ratio is 0.11. The prediction is not too far off from what we observed (0.17; first row in column F). Because the regression analysis has already been conducted, however, we know the prediction could be better. Note that one could, of course, select values of a and b such that the prediction at one x value exactly equaled the obtained y value, but the key is to bring the fitted line as close as possible to all the data points.

The vertical distances or deviations between the calculated predictions and the obtained data points are called *residuals*. We do not show all of the calculations here, but there will be residuals for all of the data points; for our single data point the residual is $0.17 - 0.11 = 0.06$. At first, given the initial parameter values, the residuals will be large. The algorithm has not adjusted the parameter values to make these distances smaller, which is the same thing as making the line fall closer to the data. How does the algorithm make these distances smaller? Whether the distances are positive or negative does not

matter; one can square the residuals and then sum them. (There is another important reason to square the residuals, which we come back to this later in the Rationale for Squaring the Sum of Squared Deviations in Regression Analysis section). We want this sum of squared residuals to be as small as possible, so the algorithm seeks to minimize this sum. The procedure is as follows: The program increases or decreases the parameter values and asks, is the sum of squared residuals smaller than the last iteration? The program continues to change the values to make the sum smaller. At some point, when changes in the parameter values produce very small changes in the sum of squared residuals, the algorithm stops and reports the best-fitting parameter values. The description of Table 10.2 lists the best-fitting parameter values obtained via least-squares regression (which is called *least squares* because of the process of minimizing the sum-of-squared deviations between the line and the data). In column G, we show the predicted data points, labeled Y' . Equation 5, the generalized matching equation, predicts these data points—they can be calculated by hand by inserting the slope and intercept values given in the table description and any one of the log reinforcer ratios in Table 10.2. The deviation between the obtained data, Y , and the predicted data, Y' , is shown in column H. If each of these values is squared, we obtain the squared deviations, shown in column I. The sum of these deviations in column I is what the algorithm minimizes to obtain the best-fitting parameter values.

To show the accuracy of the regression model visually, vertical lines can be drawn between each data point and the predicted line (these lines represent the residuals), which is shown in the left panel of Figure 10.5. The distances are very small. In linear (and nonlinear) regression, these distances are compared with the distances given by another model, the null hypothesis. The null model asserts that the mean of the obtained data can account for (predict) the results better than the more complicated two-parameter model. The null model is more parsimonious. There are no parameters, just the calculated mean of the data. The accuracy of this model is shown in the right panel of Figure 10.5. To quantitatively characterize this visual comparison

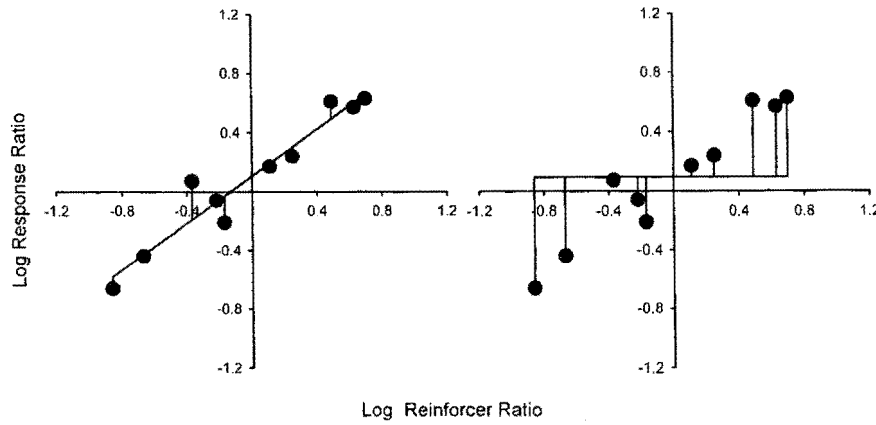


FIGURE 10.5. A visual representation of how linear regression is a matter of model comparison. The plot on the left shows the regression model, the alternative hypothesis. The plot on the right shows the null model. In linear regression, one compares the vertical distances between the data points and the line on the left to the vertical distances between the data points and the horizontal line on the right.

between the two models, sum the squared residuals in the regression model (denoted SS_{REG}), and sum the squared deviations in the null model (denoted SS_{TOT} ; TOT stands for total deviation from the mean). The ratio of these quantities is subtracted from 1, which provides the first index of whether the model is accurate:

$$r^2 = 1 - \left(\frac{SS_{REG}}{SS_{TOT}} \right) \quad (6)$$

The r^2 resulting from the linear regression analysis is shown in Table 10.2. If the value of SS_{REG} is close to SS_{TOT} , then the value of the ratio will be close to 1, which will result in an r^2 close to 0. If SS_{REG} is very small compared with SS_{TOT} , the ratio will be small and the r^2 value will be close to 1. Because r^2 is a ratio, it can be negative when the null model predicts the data better than the regression model. In other words, a negative value is obtained when SS_{REG} is larger than SS_{TOT} . Some authors designate the r^2 in different ways, such as by reporting the variance accounted for in terms of a percentage (r^2 multiplied by 100), abbreviated VAF or VAC. Whether r^2 , variance accounted for, or even variance explained, all of these represent a relative comparison of distances between the data and the predicted line (or curve; see the section Evaluating Nonlinear Models: Nonlinear Regression later in this chapter)

and the distances between the data and a horizontal line at the mean. In other words, r^2 is a ratio of how well the model predicts the data relative to the simpler null model.

The r^2 is just the first step in evaluating the model's accuracy. Examine the left panel in Figure 10.5, particularly the patterns of the data points around the line. One can see that about the same number of data points are above or below the line, that the data points are approximately the same distance from the line, and that the deviations, positive or negative, are not systematically related to the x-axis values, a condition known as homoscedasticity. Violations of homoscedasticity, a condition known as heteroscedasticity, means that there is a serious problem with the data or the model, even if one has a high r^2 . To understand why, examine Figure 10.6. The left and middle panels show different examples of heteroscedasticity. In the left panel, the model (the line) obviously and systematically misses the data. When the log reinforcer ratio is negative, the line overpredicts and then underpredicts the data. When the log reinforcer ratio is positive, the line tends to overpredict the data. The systematic pattern of deviations violates the requirement of homoscedasticity. In the middle panel, it is harder to see how the data depart from homoscedasticity, which is often the case in quantitative modeling. Although the r^2 may be high, it does not tell the whole story. Careful visual

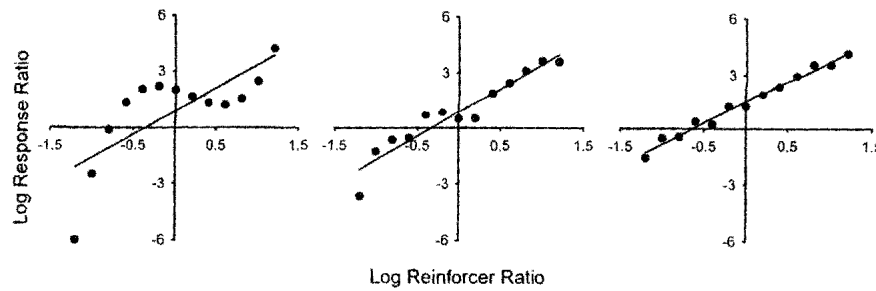


FIGURE 10.6. Two examples of fits of a linear equation to data showing heteroscedasticity (left and middle panels) or homoscedasticity (right panel). The left panel shows that the model underpredicts the data when reinforcer ratios are smaller. The middle panel shows a more subtle example, and it may be difficult to see heteroscedasticity. The right panel shows what appears to be a homoscedastic pattern, in which the data are evenly spread along the range of the fit.

inspection reveals that more data points are above the line and that within a certain range of the x-axis, approximately -0.8 to -0.2 , all the data points fall above the line. Finally, in the right panel, one can see that the number of points above and below the line are approximately equal, that the deviations above and below are roughly equal, and that there is no systematic relation between positive and negative deviations and x value, in keeping with the requirement of homoscedasticity. To be sure whether the model produces systematic departures from homoscedasticity, one needs to perform a finer grained analysis, which is called *residual analysis*. Most programs routinely report residual analyses, and we recommend careful inspection and reporting of these results.

Residual analysis can be performed in several ways. All of them use the same general procedure: Evaluate the pattern of residuals as a function of the independent variable or as a function of the predicted data (e.g., each Y' in column G of Table 10.2) to determine whether the residuals are evenly and randomly distributed. For example, one of the most basic ways to perform residual analysis is to plot the raw residuals (i.e., the difference between the predicted and obtained values) as a function of the independent variable. The residuals from Figure 10.6 are plotted in Figure 10.7 as a function of their corresponding reinforcer ratios. The order of the panels corresponds to the panels in Figure 10.6. Thus, the left panel shows obvious departures from homoscedasticity. The middle panel in Figure 10.7

shows more clearly than that of Figure 10.6 that the data are heteroscedastic. The right panel shows a desirable pattern of residuals, one that shows homoscedasticity. To perform a residual analysis using the data from Table 10.2, plot the data in column H, the raw residuals, as a function of the independent variable shown in column E, the logged reinforcer ratios.

Residual analysis can be performed in other ways (Cook & Weisberg, 1982). The standardized residual could be plotted as a function of the independent variable. The standardized residual is calculated by dividing each raw residual by the standard deviation of residuals. Standardized residuals allow one to more easily detect the presence of outliers (e.g., a residual ± 2 standard deviations from the mean of the residuals). To detect patterns in the residuals, one could use inferential statistics to supplement visual analysis. For example, in some cases a correlation between the residual and the independent variable could be calculated (e.g., Dallery, Soto, & McDowell, 2005; McDowell, 2005). A significant correlation would indicate systematic departures from homoscedasticity, and a nonsignificant correlation (which is desirable) indicates homoscedasticity. There are more advanced techniques to detect patterns in the residuals, which involve fitting curves to the residuals. Some authors use cubic, quadratic, and other polynomial equations to detect these deviations from homoscedasticity (e.g., see Sutton, Grace, McLean, & Baum, 2008). Additionally, there are techniques that involve examining the number

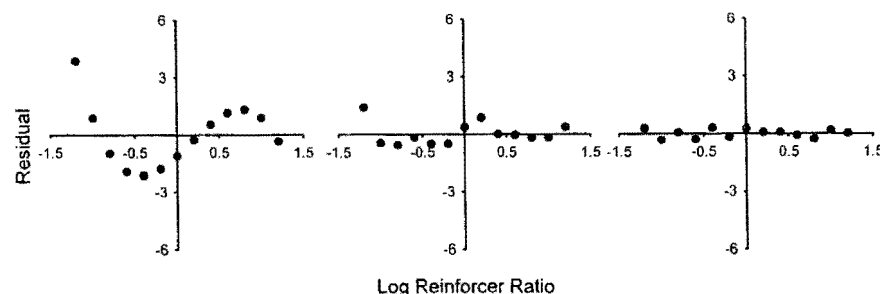


FIGURE 10.7. Residual plots of data shown in Figure 10.6. The order of panels reflects the order in Figure 10.6. See text for further details.

of residuals that have a positive or negative sign and the number of consecutive residuals of the same sign, called *runs of residuals* (McDowell, 2004; Motulsky & Christopoulos, 2004). Given the logic of residual analysis, the latter runs test should show that an equal number of residuals are above and below the line and that there are no consecutive runs of one sign or the other.

There are several potential consequences if non-random residuals are obtained. First, it may mean that the assumption of random scatter is violated and that using an ordinary least-squares regression technique may not be appropriate. We explain why this is the case in the Rationale for Squaring the Deviations in Regression Analysis section. Second, the equation might be incorrect. Perhaps the equation form is incorrect, a parameter is missing from the equation, or the parameters have not been adjusted properly (see Motulsky & Christopoulos, 2004, on troubleshooting bad fits). Something may also possibly have gone wrong with the experiment. Perhaps some uncontrolled variable was responsible for the systematic departures, or an error was made in transcription or data analysis. Before dismissing a model because of nonrandom residuals, one should be sure that an error in the experiment or analysis is not the source of the nonrandom residuals.

Although residual analysis is critical in evaluating the accuracy and validity of a particular model, for example, the generalized matching equation, residual analysis can also be used to compare models. Sutton et al. (2008) used several datasets to distinguish two models of choice and performed what might be called a meta-analysis of residuals. Because residuals can show deviations at a finer grained level

than overall measures of model accuracy such as r^2 , they allow researchers to more easily evaluate model accuracy (see also Dallery et al., 2005). Sutton et al. concluded that although the r^2 s of the two models were equivalent, analysis of the residuals yielded subtle, yet important distinctions between the models.

Thus far, we have considered three essential steps in performing linear regression: careful visual inspection of the data, calculating r^2 , and performing residual analysis. Researchers must be sure that a line describes the data. If it does not, they may need to evaluate a different model, but only if the alternative model makes theoretical sense when applied to the data. This issue requires critical analysis. There are almost always theoretical reasons for choosing a particular model, and thus regression analysis in particular and quantitative modeling in general are not simply a matter of obtaining a good fit.

ADDITIONAL QUESTIONS WHEN EVALUATING LINEAR REGRESSION

Do the Parameter Values Make Sense?

Estimated parameter values must be plausible, both in terms of precedent and in relation to the dataset to which the equation is fitted. For example, in the case of linear regression, the y-intercept should be plausible given the obtained data in the experiment. What should one expect to see if the value of the independent variable is 0? In the case of generalized matching, the independent variable is 0 when the reinforcer rates are equal. Using the example from the preceding section, assume each confederate delivered 60 reinforcers per hour; thus, $60/60$ is 1, and the log of 1 is 0. A safe initial assumption,

therefore, is that one would observe equal response rates on the two alternatives (i.e., a log response ratio close to 0), and some deviation might be expected if the response alternatives differ in some way (e.g., one confederate is more genuine or the quality of feedback is different). That is, one might expect some bias. If the obtained parameter value is widely discrepant from the existing literature, then something is most likely wrong. Solving the discrepancy may require some detective work. Perhaps the problem is lack of experimental control or human or measurement error, or some variable other than reinforcer rate is exerting an effect.

Is There Error in the Values of the Independent Variable?

When researchers perform regression analysis, they are predicting the values of the dependent variable at specific values of the independent variable, which means that they need to obtain precise numbers for their obtained data. In our earlier example, this would mean precise reinforcer rates. For example, if one uses an averaged value of 6.0 reinforcers per hour as one of the independent variable values in linear regression, one must be sure that the reinforcer rates going into this average are extremely close to 6.0 reinforcers per hour. If the measurement of a particular value of the independent variable has a lot of error or variability, one cannot make precise predictions. Regression analysis assumes that there is no, or very little, error variance in the independent variable. In the case of a generalized matching analysis, there will be some variability in the independent variable, the obtained log reinforcer ratio, because the obtained log reinforcer ratio is plotted and depends in part on the rate of responding at each alternative. (One could plot the experimentally programmed reinforcer ratio, which has no error, but the programmed ratio may not be what the animal experiences.) Because the schedules for each alternative are VI schedules, the variability in reinforcer rates will be minimal compared with the variability in response rates, and thus the resulting reinforcer ratio will have minimal variability. For example, under a VI 60-second schedule, a pigeon could press once a minute or 100 times a minute and receive about the same number of reinforcers.

Therefore, the amount of variability is not enough to invalidate the use of ordinary linear regression techniques. If the variability in the independent variable is high, alternative regression approaches are available (Brace, 1977; Riggs, Guarnieri, & Addelman, 1978; although the same considerations apply for nonlinear regression, these approaches apply only to linear regression). To our knowledge, there are no specified rules about how much variability is too much to warrant an alternative approach.

Is Another Model More Appropriate?

Even when the model accounts for the variability in the data points (i.e., a high r^2), the residuals are homoscedastic, and the parameter values are sensible, there may be other questions regarding the model that cannot be answered by linear regression analysis. For example, one may ask whether another model of concurrent choice can account for more of the variance in the data. In the case of concurrent schedules of reinforcement, in addition to the generalized matching law, alternative models can account for choice (e.g., Dallery, McDowell, & Soto, 2004; McDowell & Kessel, 1979). One may also ask whether a more parsimonious model can account for the same amount of variance. In general, a more parsimonious model would be one that contained fewer parameters. To our knowledge, no current model can account for the data and is more parsimonious than generalized matching. In other areas of research, however, questions about parsimony and accuracy are still quite relevant (e.g., Mazur, 2001; McKerchar et al., 2009). Questions about accuracy and parsimony can be addressed using model comparison approaches, which we discuss later (see also Mazur, 2001, for an example of model comparison).

The underlying assumptions of the selected model should also be examined. Indeed, a highly accurate model may be based on faulty assumptions. There are no tidy rules for discovering these assumptions; it takes critical thinking and a thorough understanding of the mathematical model. One assumption of generalized matching, for example, is that sensitivity to reinforcer rates (i.e., a in Equation 5) should be constant for an individual subject across changes in reinforcer magnitude or quality. By constant, we mean that a should not

change if, for instance, a is estimated using small food pellets or large food pellets. If sensitivity to reinforcer rates does change when different reinforcer magnitudes are used, then the assumptions underlying the generalized matching equation have been violated, and the application of the equation is invalid. At least one study has shown that sensitivity to reinforcer rates does change depending on reinforcer magnitude (Elliffe, Davison, & Landon, 2008). Elliffe et al.'s (2008) findings question a matching-based understanding of how multiple independent variables—reinforcer rate and magnitude—interact and govern choice. Whether these findings will be confirmed by future research is an important issue, and such work may inspire new models that will yield new insights into the determinants of choice.

RATIONALE FOR SQUARING THE DEVIATIONS IN REGRESSION ANALYSIS

The main feature of the regression analysis is minimizing the sum-of-squared deviations. The story behind the reason researchers seek to minimize the squared deviations, or perform so-called least-squares regression, is an interesting one. In 1801, at the age of 23, Carl Friedrich Gauss used the method of least squares to predict, with unprecedented precision, the path of a dwarf planet named Ceres. He made about 22 observations over a 40-day period before Ceres went behind the sun. To predict where it would reappear almost half a year later, he developed the method of least squares. The particular equation he used to predict the path of the planet need not concern us. What made the method work was Gauss's assumption that any error in the observations followed a normal distribution. To illustrate why, we provide a more familiar example than planetary motion.

Consider one of the more highly controlled experimental preparations in behavioral science: a pigeon pecking a lighted key for brief access to grain. After every 20 pecks, access is granted for a brief period. As response rate is measured from session to session, the numbers will vary. One day the response rate is higher, the next it is lower. However, all of the numbers revolve around some average. The distribution of the numbers is the

key—the scatter of response rates is normally distributed, or Gaussian. Most of the numbers will be close to the average, but some scatter will also be observed further away from the average. In other words, scatter close to the mean is more likely than scatter far away from the mean.

Assume the average is known because the pigeon has been observed for 100 consecutive days. The average is 50 responses per minute. On Days 101 and 102, one student takes measurements, and on Days 103 and 104, another student takes measurements. The first student obtains 45 and 55, and the second obtains 49 and 59. What if the same students made two more measurements each? Which pair of observations would be more likely the second time around? This is a question about predicting behavior and how to maximize accuracy. To quantify accuracy, calculate deviations from the mean. Smaller deviations mean better accuracy. What if the researcher just calculates the deviations from the mean for both pairs? Ten units would be obtained for both pairs of observations ($5 + 5 = 10$ and $1 + 9 = 10$). The researcher would conclude that both pairs are equally as likely because they have equal total deviations from the mean. However, these calculations do not take into account a property of the deviations around the mean: the normal, Gaussian distribution. To take this property into account, the deviations are squared, that is, $5^2 + 5^2 = 50$ and $1^2 + 9^2 = 82$. On the basis of these calculations, the first pair of observations is more likely (see also Motulsky & Christopoulos, 2004).

The reason that squaring the deviations produces the most accurate predictions can be seen in Figure 10.8. The abscissa shows the values of the observations, and the ordinate shows the frequency of their occurrence. Consistent with the example, behavior has been measured for 100 days. On the basis of the distribution, one can see how the observations made by Student A are more likely; the likelihood of Student A's observations of 45 and 55 is greater than that of the observations made by Student B.

If the observational error in a system under study shows a Gaussian distribution, whether planetary motion or a pigeon's pecking, then least-squares regression will be most accurate in generating the best parameter values, the values that make the best

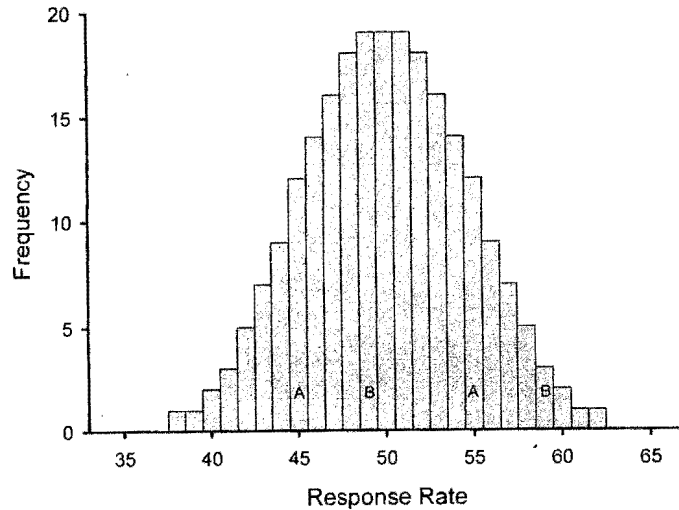


FIGURE 10.8. An example of a Gaussian distribution. The graph shows the frequency of observations, in this case response rates, over the course of 100 sessions. The distribution has a mean of 50. Two pairs of observations are taken, one by Student A (indicated by the As in the graph) and two by Student B (indicated by the Bs in the graph).

predictions. The error could be intrinsic to the system itself, or perhaps the variation is the result of some extrinsic factor such as measurement error or how much the experimenter feeds the pigeon between sessions. Regardless of the source, the observations over time, such as response rates over the course of 100 sessions, should show a Gaussian distribution. It is important to understand how and why the sum-of-squared deviations incorporate the Gaussian distribution in making predictions.

Other methods to perform regression do not assume a Gaussian distribution. Usually, least-squares regression is a justified practice to assess a model, but there are times when the distribution of observations do not show Gaussian scatter; sometimes the scatter of observations increases as the value of the independent variable increases, perhaps the scatter is wider than a Gaussian distribution, or perhaps the distribution is skewed and therefore non-Gaussian. For example, distributions of interresponse times are often skewed, and therefore using linear regression on mean interresponse times would not be appropriate in these cases. Techniques are available to perform regression on the basis of these distributions that involve different methods of weighting the residuals when calculating the best fit.

We do not present these techniques because of space limitations (see Motulsky & Christopoulos, 2004), but we do want to emphasize why least-squares regression would not be appropriate for these distributions. In a nutshell, it is because some residuals will have undue influence on the fit (similar to the effect of an extremely large or small value on the average of a set of values). The logic also explains why outliers require some special consideration in regression, even if one is confident that the scatter of observations follows a Gaussian distribution. Because the deviation to the outlier will be large, it will have a large impact when one attempts to fit the curve (or line). Consequently, the outliers will have a disproportionate impact on the values of the parameters, which would be unfortunate if the outlier was a fluke observation (e.g., the food dispenser stopped working one day and so did the animal). There are methods to reduce the influence of outliers, and these methods are called robust least-squares regression. Computer programs will differ in which robust methods are offered, so we advise a search for these robust methods if outliers are present (and the reason for the outlier has been critically analyzed and including the outlier in the regression analysis is justified).

AN EXTENSION OF MATCHING THEORY'S QUANTITATIVE DESCRIPTION OF CHOICE

Just as the development of linear models often begins with data, so too does the development of a nonlinear model. For example, imagine we have arranged a laboratory situation in which a human has one person to talk to, not two, and the rates of reinforcer delivery for talking about a certain topic, say politics, are varied across conditions. In this case, reinforcers consist of brief periods of eye contact (Beardsley & McDowell, 1992). Eye contact is provided according to several VI schedules of reinforcement (manipulated across conditions), but only when politics are discussed. We find that the rates of political speech increase with reinforcer rates. The relation is not a straight line, however; it is a curve. (Two examples of such curves are depicted in Figure 10.2.) At first, responding increases rapidly when reinforcer rates increase, and eventually responding levels off. The equation, of course, will reflect this hyperbolic shape in the observed data, and in that sense it does not take us beyond our observations. Our model, however, should extend our observations in several ways. Can it unify our observations of human behavior with the behavior of other organisms? Can it make novel and testable assertions about why the data take the shape they do?

Before answering these questions, let us sketch the development of the hyperbolic equation. In 1968, Catania and Reynolds published an enormous and elegant dataset on how pigeons' rate of pecking changed with changes in rate of VI reinforcement. The relation, as in our example of social dynamics, appeared hyperbolic. Thus, researchers knew that any quantitative account of reinforced responding on a single VI alternative must be consistent with a hyperbolic form. In 1970, Herrnstein demonstrated that one can obtain a hyperbolic model from the original matching equation, Equation 2. In deriving a hyperbolic equation from Equation 2, Herrnstein made several assumptions. First, he assumed that even when only one alternative is specifically arranged, other extraneous, unarranged behavioral alternatives exist. Consider the pigeon's situation in Catania and Reynolds. In addition to pecking the lighted key, the

pigeon may engage in other behavior such as walking around, wing flapping, and so on, and presumably, each behavior produces some natural reinforcer. Similarly, a human engaged in political speech could talk about a different topic, fidget, or stare into space. Therefore, a single alternative arrangement is really a concurrent schedule, in which one choice is the target alternative and the other is the aggregate of all extraneous behaviors. Responding in a single-alternative situation can be expressed as

$$\frac{R}{R + R_e} = \frac{r}{r + r_e}, \quad (7)$$

where R_e and r_e represent the rates of extraneous responding and reinforcement, respectively (see Herrnstein, 1974, for a further discussion). In other words, R_e and r_e represent any responding and reinforcement extraneous to the target alternative. The subscripts have been dropped from the target alternative because only one schedule of reinforcement is explicitly arranged.

Second, Herrnstein (1970) also assumed that R (target responding) and R_e (extraneous responding) are exhaustive of the total amount of behavior possible in a given environment. In other words, $R + R_e$ is all an animal can do in a given situation. Equation 7 can be solved for R by letting $k = R + R_e$ and then rearranging, which produces the familiar Equation 1:

$$R = k \left(\frac{r}{r + r_e} \right).$$

The parameter k is now interpreted as total behavior possible in a situation. By assuming that all behavior is choice and that the total amount of behavior in a situation is constant, Herrnstein (1970) derived Equation 1 from Equation 2.

Herrnstein's (1970) theory consists of a quantitative statement about how reinforcer rate affects response rate (i.e., hyperbolically). More important, the theory also explains why reinforcement affects behavior in the manner specified by Equation 1: Reinforcement alters the distribution of a constant amount of behavior (Herrnstein, 1970, 1974). Now, at least conceptually, one can see how Herrnstein's model links responding in situations in which only

one choice is experimentally arranged and situations in which two choices are experimentally arranged. Both represent choice situations. Indeed, Herrnstein went so far as to say that all behavior is choice. A pigeon's key pecking and a person's talking can be described by the same equation. The theory also explains why one feature of the hyperbolic shape is seen: asymptotic responding. Once enough reinforcers are delivered for some activity, the limits of the distribution of behavior are presumably reached (i.e., exclusive responding on the arranged alternative), and thus asymptotic levels of responding are observed.

EVALUATING NONLINEAR MODELS: NONLINEAR REGRESSION

The preceding conceptual analysis highlights some of the surprising and interesting consequences of quantitative models. Researchers can unify phenomena and explain why they see similar functional relations across situations and species. These consequences are moot unless the model is accurate, however, and as such we must return to analysis. Table 10.3 shows empirical observations in the single-alternative situation involving social interaction and political speech, and it follows the same general format as Table 10.2. Figure 10.9 shows a plot of the data in columns A and B, the response

and reinforcer rates, and the curve represents the least-squares fit of the hyperbolic equation.

Least-squares regression for nonlinear models works in the same way as it does for linear models. As before, start with some initial parameter values for k and r_e . Here, it may be more difficult to guess where to start for the initial values; it is easier if one examines a graph of the data. Inspect the data points in Figure 10.9. The value of the asymptote k is measured in the units of the dependent variable—usually responses per minute. Because k reflects the

TABLE 10.3

Regression Calculations for a Nonlinear Model

A: rnf/hr	B: rsp/min		C: predicted		E: $(Y - Y')^2$
	(Y)	rsp/min (Y')	D: $Y - Y'$		
180	32.0	31.46	0.54		0.2916
138.6	30.1	30.45	-0.35		0.1225
98	28.0	28.78	-0.78		0.6084
22.5	18.8	17.68	1.12		1.2544
15	16.3	14.14	2.16		4.6656
9	6.7	10.09	-3.39		11.4921

Note. Columns A and B show response and reinforcer rates, and column C shows predicted response rates calculated by Equation 1 with $k = 35.41$ and $r_e = 22.57$. Other conventions are identical to those in Table 10.1. Regression calculations: $SS_{TOT} = 478.43$, $SS_{REG} = 18.43$, $R^2 = 0.96$. rnf = reinforcer; rsp = response.

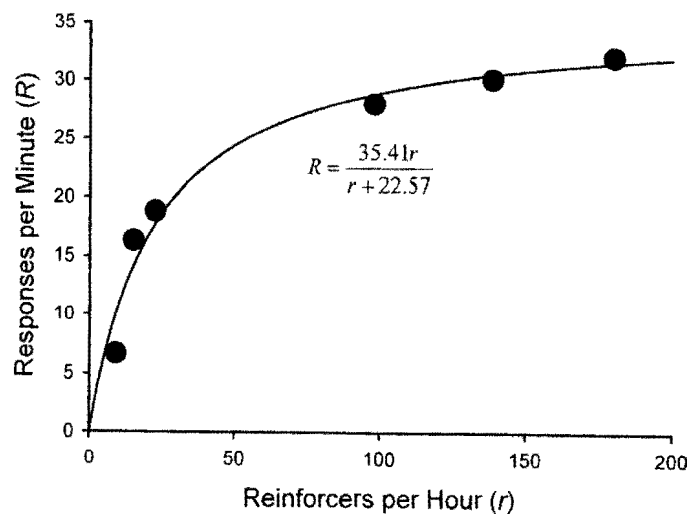


FIGURE 10.9. The curve shows Herrnstein's (1970) hyperbolic equation fitted via least-squares regression to the data in Table 10.3.

asymptote, one could start with the maximum response rate observed, or 32 responses per minute. Mathematically, the parameter r_c represents the reinforcer rate necessary to obtain half the asymptotic response rate, k . The reinforcer rate for the maximum response rate was 180 reinforcers per hour, so half of this reinforcer rate could be used as the initial value of r_c . We should note that, by definition, r_c represents the aggregate rate of reinforcement from sources extraneous to the target alternative. These extraneous reinforcers are typically not measured directly; rather, they are estimated in terms of the reinforcer delivered on the instrumental alternative. For example, if political speech were reinforced with eye contact, then an r_c of 10 would be read as 10 eye contacts per hour (the time unit for r_c is usually reinforcers per hour).

In performing nonlinear regression, having a sense of what the parameters mean allows the researcher to use the data to determine initial parameter values. Then, the computer program (or algorithm) inserts the initial estimates (32 responses per minute and 90 reinforcers per hour) into Herrnstein's (1970) equation to obtain a predicted response rate for each of the measured reinforcer rates. The computer program adjusts the parameter values to minimize the sum-of-squared residuals. Table 10.3 shows the best-fitting parameter values of k and r_c obtained via least-squares regression. The predicted data based on these values are shown in column C.

The first index of the model's accuracy is the R^2 (by convention, results of nonlinear regression are reported using R , and results of linear regression are reported using r). Similar to what was depicted in Figure 10.5, the R^2 compares the vertical distances of the data to the curve with the vertical distances of the data to a horizontal line at the mean. More specifically, a ratio of the sum-of-squared residuals to the total sum-of-squares is calculated. In our example, SS_{REG} is 18.43, and SS_{TOT} is 478.43. Subtracting the ratio of SS_{REG} from SS_{TOT} from 1 yields an R^2 of .96.

A residual analysis is also performed to assess whether there are systematic departures from homoscedasticity. The R^2 tells only part of the story. Again, one could perform several different kinds of residual analysis (see earlier), but for simplicity the

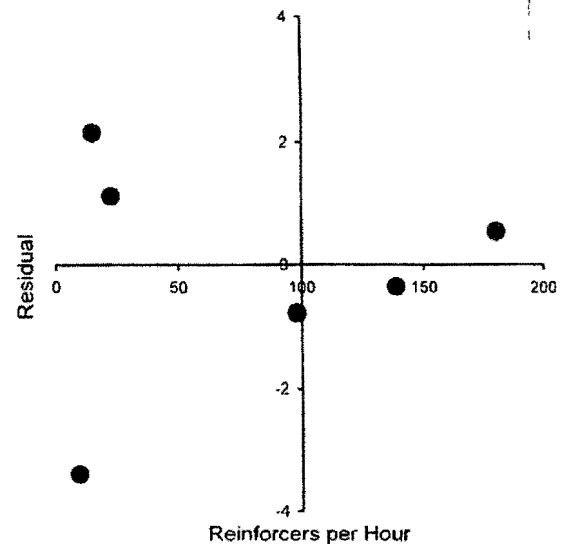


FIGURE 10.10. Plot of the residuals of the fit of Herrnstein's (1970) hyperbolic equation. The residual plot does not suggest systematic deviations for homoscedasticity.

residual is plotted as a function of the independent variable, which is shown in Figure 10.10. One can see that the scatter of residual values is relatively even across the range of reinforcer rates. More data may be needed to assess the assumptions of least-squares regression, but the analysis suggests that there are no obvious departures from the assumptions.

ADDITIONAL QUESTIONS WHEN EVALUATING NONLINEAR REGRESSION

Do the Parameters Make Sense?

To answer this question, as in linear regression, one has to know what the parameters mean and in what units they are measured. If researchers obtain an estimate of k of 1,000, and they have measured response rates in units of words per minute, this estimate of asymptotic response rates would not be feasible. Moreover, even if the maximum observed response rate were 100 or so, one should be suspicious of such a parameter value because it exceeds the observations by orders of magnitude. That is, if the parameter value exceeds what is reasonable on the basis of observed values, the researchers should think carefully about the obtained parameter value. It is indeed possible to obtain very large k s, but this

is usually because the researcher has not collected enough data under high reinforcer rates (i.e., in which response rate is close to asymptote). In choosing the VI schedules, the researchers have not sampled the parameter space effectively. For example, one would say the parameter space has not been sampled adequately if only the three lowest reinforcer rates shown in Figure 10.9 (three leftmost data points) were experimentally arranged. Even visually, using these three data points, the researchers would not be able to make a reasonable guess about the asymptote of the function. Their guess might be large (and uncertain). Similarly, the estimate produced by the regression program might be large (and uncertain, as measured by the standard error of the parameter estimate, for example; Motulsky & Christopoulos, 2004). Given that the parameter space has been adequately sampled, as shown in Figure 10.9, the obtained k of 35.41 responses per minute makes sense given the observations. The obtained r_e of 22.57 reinforcers per hour also makes sense. Although it may seem odd that extraneous reinforcers are measured in units of the delivered reinforcer, eye contacts, the value of eye contacts is plausible given the observed data.

Is There Error in the Values of the Independent Variable?

As noted for linear regression, when one performs nonlinear regression analysis, one is predicting the values of the dependent variable at specific values of the independent variable. We do not go into this issue in detail because the arguments are the same as outlined in the Evaluating Linear Models: Linear Regression section. Briefly, recall that if there is a lot of variability in the independent variable, one cannot answer with precision the question of how the dependent variable changes with changes in the independent variable.

Is Another Model More Appropriate?

As with linear models, when evaluating a nonlinear model, one should consider whether there is evidence that questions the assumptions underlying the model or evidence for other candidate models that seek to describe the same environment–behavior relation. For example, although Equation 7

typically accounts for more than 90% of the variance in response rates under VI schedules in laboratory, applied, and naturalistic settings (see de Villiers & Herrnstein, 1976; Fisher & Mazur, 1997; McDowell, 1988; and Williams, 1988, for reviews), there are some reasons to think it may not be wholly correct. As noted by Shull (1991), “The fact that an equation of particular form describes a set of data well does not mean that the assumptions that gave rise to the equation are supported” (p. 247). The assumptions underlying Herrnstein’s (1970) hyperbola have been examined in terms of how the parameters change with manipulations of environmental variables (see Dallery & Soto, 2004, for a review). According to the equation, r_e should vary directly with changes in the extraneous reinforcer rate, and this prediction has some empirical support (Belke & Heyman, 1994; Bradshaw, Szabadi, & Bevan, 1976). Other studies, however, have questioned this prediction (Bradshaw, 1977; Soto, McDowell, & Dallery, 2005; White, McLean, & Aldiss, 1986). Similarly, there are conflicting findings for the prediction of a constant k as reinforcer properties are manipulated. Recall that this is required by Herrnstein’s (1974) theory: Reinforcement alters the distribution of a constant amount of behavior among the available alternatives (de Villiers & Herrnstein, 1976, p. 1151; Herrnstein, 1974). Thus, the evidence has suggested some limitations to the assumptions underlying the hyperbola, and these limitations are independent of the criteria by which one evaluates the appropriateness of the model in describing a particular data set. Moreover, other models account for the same hyperbolic relations as Herrnstein’s model and predict the conditions under which k should remain constant or vary (Dallery, McDowell, & Lancaster, 2000; Dallery et al., 2004; McDowell, 1980; McDowell & Dallery, 1999).

There is another reason to question the validity of Herrnstein’s equation. Recall that the original, strict matching equation, on which Herrnstein’s equation is based, was modified to account for bias and insensitivity. If bias and insensitivity are ubiquitous in concurrent schedules, they should be incorporated into Herrnstein’s (1974) account of single-alternative responding. We do not develop the equation that incorporates these parameters into Herrnstein’s original equation (see McDowell, 2005,

for the derivation). As one might imagine, incorporating these parameters makes the equation appear more complicated. However, the equation describes the same hyperbolic shape, the same environment-behavior relation. Evaluating the equation involves the same steps outlined earlier. Indeed, in comparing the family of equations that make up the so-called modern theory of matching with the classic theory of matching, McDowell and colleagues (Dallery et al., 2005; McDowell, 2005) made extensive use of residual analysis to detect subtle, yet important deviations from the original, classic theory of matching. Although Dallery et al. (2005) and McDowell (2005) concluded that the modern theory represented a superior theory, alternative quantitative accounts also provide useful, general, and informative descriptions of the same environment-behavior relations (e.g., Killeen & Sitomer, 2003; Rachlin, Battalio, Kagel, & Green, 1981).

MODEL COMPARISON

Evaluating the accuracy and generality of a single quantitative theory is an important first step in describing environment-behavior relations. Comparing the model with other accounts is also a critical feature of the scientific process. Indeed, making distinctions between models can help further isolate controlling variables, develop critical empirical tests to distinguish the models (Mazur, 2001), and potentially lead to new insights about the processes responsible for behavior change in applied settings (Nevin & Grace, 2000). In the next section, we present two techniques to compare models. One technique, the extra sum-of-squares *F* test (Motulsky & Christopoulos, 2004), is based on conventional inferential statistics, and the other technique, Akaike's information criterion (AIC; Akaike, 1974; Motulsky & Christopoulos, 2004), is based primarily on information theory (Burnham & Anderson, 2002). The objective of each technique is to provide a quantitative basis on which to judge whether one model provides a better account of the data than does another. Both methods evaluate goodness of fit relative to model complexity (i.e., the number of parameters). This is because the more parameters a model has, the better it will describe the data, and

thus any improvement in goodness of fit relative to another model must be considered in the context of this increase in the number of parameters.

Before comparing models using either the sum-of-squares *F* test or AIC, researchers must determine whether both models provide a reasonable and accurate account of the data: (a) Do they describe the data well (i.e., high r^2 or R^2), (b) are the residuals randomly distributed, and (c) are the parameter values sensible? If the answer to each of these three questions is yes, then researchers can proceed to model comparison with the *F* test or AIC. If the answer to one of the questions is no, then researchers must either resolve the issue (e.g., if nonsensible parameters are the result of experimental error) or choose between the models on the basis of the residuals or goodness of fit (i.e., if one model has nonrandom residuals or does a poor job of describing the data, then choose the other model). In addition, researchers must ensure that all datasets are expressed in the same units. They need to compare apples to apples. If the dependent variable in one data set is expressed in responses per second and in another it is expressed in responses per minute, the sum-of-squared deviations will be much larger in the former than in the latter. A comparison with different units is not appropriate. The researcher must either reexpress the data so that the units are identical (e.g., multiply or divide by 60 in the previous example) or find a way to standardize, or normalize, the data before making the comparison.

The appropriate model comparison test depends on whether the models are nested. Models are nested when one model is a simpler case of the other model. For instance, the original (Equation 2) and generalized matching (Equation 5) equations are nested models because Equation 4 can be reduced to Equation 2 when a and b are equal to 1. If the models are nested, they can be compared with an extra sum-of-squares *F* test or with the AIC. If they are not nested, they can only be compared using AIC (or an alternative statistical approach using parametric or nonparametric methods).

Extra Sum-of-Squares *F* Test to Compare Nested Models

Let us use strict matching and generalized matching as an example of model comparison using the data in

Table 10.2. Perfect matching is described by Equation 5 with a and b set to 1. Generalized matching is also described by Equation 5, but the parameters are free to vary. Assume that both models describe the data well, have randomly distributed residuals, and have sensible parameter values. The logic of the extra sum-of-squares F test is relatively simple. Goodness-of-fit is being compared along with the complexity of each model. The goodness-of-fit is measured by the sum-of-squared residuals. The complexity is measured by the number of parameters in each model. More parameters that are free to vary mean more complexity. The rule of parsimony asserts that simpler models should be preferred, so any increase in complexity should be more than offset by an increase in accuracy. Before going into the specific calculations, the F ratio compares accuracy (sum-of-squares) and complexity (parameters). If the calculated F ratio is high enough (and the corresponding p value is low enough), then the more complicated model can be accepted (generalized matching, in which the parameters are free to vary). If not, the simpler model is accepted (strict matching, in which the parameters are not free to vary).

To calculate the extra sum-of-squares F test, one begins by calculating the sum-of-squared residuals for each model. The sum-of-squared residuals for strict matching in this case is 0.34; for generalized matching, it is 0.13 (see Table 10.4). A smaller sum-of-squares is expected for generalized matching, and in general for models with more parameters that are free to vary. Next, the degrees of freedom for each model are calculated as the number of data points minus the number of parameters. For strict matching, the degrees of freedom are 10 (10 data points minus 0 free parameters); for generalized matching, the degrees of freedom are 8 (10 data points minus 2 parameters). Now, the relative difference in sums-of-squares and the relative difference in degrees of freedom are calculated. The relative difference in sums-of-squares is calculated by subtracting the alternative hypothesis sum-of-squares from the simpler hypothesis' sum-of-squares, and then dividing that difference by the alternative hypothesis

TABLE 10.4

Obtained Logged Reinforcer and Response Ratios and Predicted Log Response Ratios and Corresponding Regression Calculations for Strict Matching and Generalized Matching

A: log rnf ratio	B: log rsp ratio	C: predicted log rsp ratio from strict matching	D: predicted log rsp ratio from general- ized matching
0.11	0.17	0.11	0.19
-0.37	0.07	-0.37	-0.19
0.49	0.61	0.49	0.49
-0.86	-0.66	-0.86	-0.58
0.70	0.63	0.70	0.66
-0.17	-0.21	-0.17	-0.03
0.63	0.57	0.63	0.60
-0.22	-0.06	-0.22	-0.07
0.25	0.24	0.25	0.30
-0.67	-0.44	-0.67	-0.43
Regression calculations			
SS_{TOT}	1.78		
SS_{REG}		0.34	0.13
r^2		0.81	0.93

Note. Columns A and B show the logged response and reinforcer ratios, and columns C and D show the predicted log response ratios from strict matching (Equation 5 with a and b equal to 1) and generalized matching (Equation 5) with $a = 0.80$ and $\log b = 0.10$ and their respective regression calculations. rnf = reinforcer; rsp = response.

sum-of-squares. Similarly, the relative difference in degrees of freedom is calculated by subtracting the alternative hypothesis degrees of freedom from the simpler hypothesis degrees of freedom, and then dividing that difference by the alternative hypothesis degrees of freedom. These numbers are summarized in Table 10.5. The relative difference in the sums-of-squares is 1.62 ($[0.34 - 0.13]/0.13$) and the relative difference in the degrees of freedom is 0.25 ($[10 - 8]/8$). The F ratio is the ratio of the relative difference in the sum-of-squares to the relative difference in the degrees of freedom, and it is a measure of how much better the alternative model describes the data

*In model comparison, we prefer the term *simpler hypothesis* to *null hypothesis*. We agree with Rodgers (2010) that the term *null hypothesis* has too much baggage and often refers to no effect or to chance processes. The modeling approach does not posit a null (or nil) effect; rather, it entails a specific, but simpler, hypothesis.

TABLE 10.5

Calculations Involved in the Extra Sum-of-Squares *F* Test for Model Comparison of the Descriptions of Strict Matching and Generalized Matching Accounts of the Data in Table 10.4

Model	<i>SS</i>	<i>df</i>
Simpler hypothesis (strict matching)	0.34	10.00
Alternative hypothesis (generalized matching)	0.13	8.00
Difference	0.21	2.00
Relative difference	1.62	0.25
Ratio (<i>F</i>)	6.48	
<i>p</i>	.02	

Note. Strict matching = Equation 5 with *a* and *b* equal to 1; generalized matching = Equation 5.

relative to how many more degrees of freedom the alternative model uses. In this case, the value is 6.48 (1.62/0.25) and the associated *p* value is .02. The *p* value can be found in an *F* ratio statistics table or by using an online calculator. Remember to use the correct degrees of freedom when looking up or calculating a *p* value. In this case, the degree of freedom for the numerator is 2 (the difference in degrees of freedom of our simpler and alternative hypotheses, 10 and 8, respectively), and the degree of freedom for the denominator is 8 (the alternative hypothesis degree of freedom). Typically, the criterion alpha value is set at .05, so in this case, one would reject the simpler hypothesis that the simpler model, strict matching, provides a better account of the data.

Akaike's Information Criterion to Compare Nested Models

Another technique used for comparing models is AIC. As noted earlier, AIC is based on information theory. The basis of the AIC calculation is beyond the scope of this chapter (for a brief discussion of the basis of AIC, see Motulsky & Christopoulos, 2004; for a more detailed discussion, see Burnham & Anderson, 2002). The AIC calculation allows one to determine which model is more likely to be correct (the one with the lower AIC value) and how much more likely it is to be correct. Unlike the extra sum-of-squares *F* test, AIC can be used to compare nested

or nonnested models. We use our previous example to illustrate the use of AIC for model comparison. The logic is the same as in the extra sum-of-squares *F* test: Accuracy and complexity are compared. Increases in complexity must be offset by increases in accuracy. The first step in using AIC for model comparison is to calculate the AIC for each model using the equation

$$AIC = N \ln \left(\frac{SS_{REG}}{N} \right) + 2K, \quad (8)$$

where *N* is the number of data points, *K* is the number of parameters in the model plus 1, and *SS*_{REG} is the model's residual sum-of-squares. Consider Equation 8. The AIC value depends both on how well the model describes the data (how small the residual sum-of-squares is; *SS*_{REG}) and on how many parameters the model has (*K*). When comparing two models, the model more likely to be correct is the one with the lower AIC value because a lower AIC value represents a better balance between goodness of fit and number of parameters. When the number of data points is small relative to the number of parameters, which may often be the case in behavioral experiments, AIC values will be small and should be corrected (Motulsky & Christopoulos, 2004). The corrected AIC formula is

$$AIC_c = AIC + \frac{2K(K+1)}{N-K-1}. \quad (9)$$

In the previous example, the residual sum-of-squares was 0.34 for strict matching and 0.13 for generalized matching. The number of data points is the same for each model (10), and *K* is 1 for strict matching and 3 for generalized matching. Thus, one AIC value for each model can be calculated. The corrected AIC for strict matching is -31.31, and the AIC for generalized matching is -33.43. Because the AIC value for generalized matching (designated as AIC_B) is lower than the AIC value for strict matching (designated as AIC_A), one can say that generalized matching is more likely to be correct. Some authors just present the individual AICs to make comparisons between models; the model with a smaller AIC is preferred. Another approach is to

calculate the probability that one model is more likely by using the following equation:

$$\text{Probability} = \frac{e^{-0.5(AIC_B - AIC_A)}}{1 + e^{-0.5(AIC_B - AIC_A)}} \quad (10)$$

If the corrected AIC for generalized matching, AIC_B (−33.43), and the corrected AIC for strict matching, AIC_A (−31.31), are substituted into Equation 10, the probability that generalized matching is correct is found to be approximately .74. Alternatively, it can be said that the chance that generalized matching is correct is 74%. Note, the difference could be calculated the other way and would result in a probability of approximately .26 that strict matching is correct relative to generalized matching. Finally, an evidence ratio can be calculated that quantifies how much more likely one model is to be correct relative to another model by dividing the probability that one model is correct by the probability that the other model is correct. For the current example, the probability that generalized matching is correct (.74) could be divided by the probability that strict matching is correct (.26), resulting in an evidence ratio of 2.85, which means that generalized matching is 2.85 times more likely to be correct than strict matching. These calculations are summarized in Table 10.6.

Akaike's Information Criterion to Compare Nonnested Models

When comparing models that are not nested, when neither model is a simpler version of the other model, researchers must use AIC. They must scrutinize each model (i.e., r^2 , residuals, parameters)

before subjecting them to the AIC. The sum-of-squares F test cannot be used with nonnested models. We do not provide an example of using the AIC with nonnested models because the calculations described in the previous section are exactly the same whether AIC is used for comparing nested or nonnested models. When evaluating the probability that the more likely model is correct relative to another model (Equation 10), subtract the AIC for the less likely model (i.e., the model with the larger AIC value) from the AIC for the more likely model (i.e., the model with the smaller AIC value).

One area in which AIC could be applied to model comparison is in research on temporal discounting. Temporal discounting equations describe the rate at which a reinforcer loses value as the delay to its receipt increases (Critchfield & Kollins, 2001; Mazur, 1987). Several nonnested quantitative models (e.g., exponential decay, hyperbolic decay) have been proposed that account for high proportions of the variance in responding (for a recent comparison of four discounting models, see McKerchar et al., 2009). Assuming that residual analysis does not reveal departures from randomness, nonnested AIC analysis may be useful in making comparisons between these models.

Model Comparison to Evaluate Whether an Intervention Changed Behavior

The model comparison approach can also be used to assess whether some intervention (e.g., an environmental or pharmacological intervention) produced a significant change in behavior. For example, the model comparison could be used to evaluate whether a drug or some other intervention produced

TABLE 10.6

Calculations Involved in Akaike's Information Criterion for Comparison of Strict (Equation 5 With A and B Equal to 1) and Generalized Matching (Equation 5) Accounts of the Data in Table 4

Model	SS_{REG}	N	Parameters	Akaike's	Probability	Evidence ratio
				information criterion		
Simpler hypothesis (strict matching)	0.34	10	0	−31.31	.26	
Alternative hypothesis (generalized matching)	0.13	10	2	−33.43	.74	2.85

a change in the dose–response curve in a behavioral pharmacology experiment. Similarly, one could ask whether a curve, say a demand curve (which relates consumption to response requirement), changed as a function of a manipulation such as income (number of reinforcers available per day) or degree of food deprivation. Here, the comparison is not a change in the value of a single data point, as in an analysis of variance comparing means; the comparison is between two models (i.e., equations) of the relation across many data points (e.g., comparing a fit of the generalized matching law under control and drug conditions).

One could also assess whether a particular parameter value changed as a result of some intervention. For example, one could evaluate whether the sensitivity parameter in the generalized matching equation varies as a function of some manipulation such as drug administration. Similarly, researchers in behavioral pharmacology may be interested in whether a drug alters a parameter value in a quantitative model (e.g., Pitts & Febbo, 2004). Such changes may indicate that certain behavioral mechanisms (e.g., changes in sensitivity to reinforcer amount or delay) are responsible for a particular drug effect (e.g., an increase in impulsive choice). In the interest of brevity, we provide a detailed example of model comparison and only the general logic and procedure for parameter comparison.

The central question in model comparison is, did an intervention produce a significant change in the function describing the relation between the independent and dependent variables? Consider a case in which the researcher has collected two datasets, one obtained before and one obtained after an intervention. The model comparison approach asks whether separate fits of an equation to the data, pre- and post-intervention, for example, are better than a single fit of the equation to both datasets. If the separate fits are not better, it suggests that the intervention had no effect on behavior. Note that the basic logic of all model comparison is entailed by this procedure: A more complicated model (one equation fitted separately to each dataset) is being compared with a more parsimonious model (one equation fitted to both datasets). An example of this comparison is shown graphically in Figure 10.11. The left panel shows the alternative hypothesis, and the right panel shows the simpler hypothesis (more details about the figure are discussed later). Either the F test or the AIC can be used to determine whether a particular intervention changed behavior, but we present only the F test. (For the AIC, one would need to calculate the sum-of-squared residuals, N and K , for each model. The calculations for the AIC were given earlier.) Motulsky and Christopoulos (2004) provided a succinct discussion of some factors that may lead a researcher to favor one test over the other.

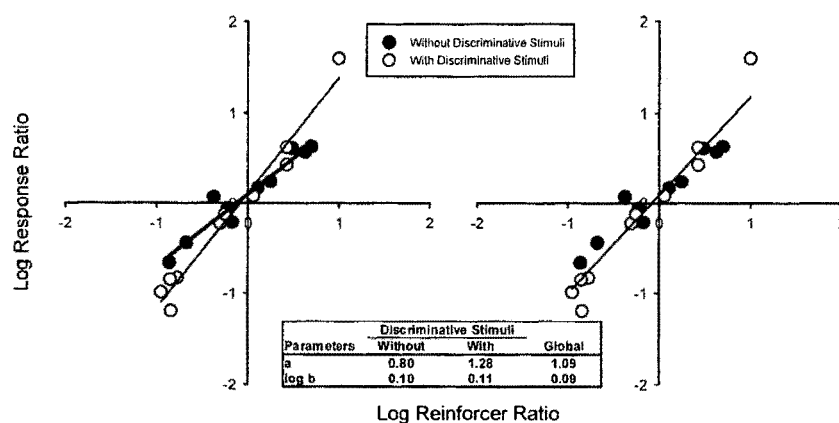


FIGURE 10.11. A model comparison approach for nested models. The left panel shows the more complicated alternative hypothesis. The alternative hypothesis is that separate fits of the equation provide a better account of the data. The more parsimonious, simpler hypothesis, shown in the right panel, is that one equation can account for all of the data. The data points represent the logged response and reinforcer ratios presented in Table 10.6.

In this example, assume we have conducted an experiment with pigeons in which we have manipulated the presence of discriminative stimuli. The discriminative stimuli, such as different colored lights, signal the availability of reinforcement at each of the two alternatives. If behavior is more sensitive to changes in reinforcer rates when discriminative stimuli are used, we can make a prediction: A fit of the generalized matching equation with discriminative stimuli will differ from the fit of the generalized matching equation without discriminative stimuli.

The first dataset, obtained without discriminative stimuli, and the second dataset, obtained with discriminative stimuli, are presented in Table 10.7 and plotted in the left and right panels of Figure 10.11. The simpler hypothesis is that the discriminative stimuli have no effect on behavior and that a single fit describes all the data. Thus, we fit the generalized matching equation to all the data pooled together, which is also called *global fitting*. The global fit is indicated by the solid line in the right panel of Figure 10.11. The values of the parameters for global fit are $a = 1.09$ and $\log b = 0.09$. The alternative hypothesis is that the discriminative stimuli did indeed change behavior and that separate fits of the generalized matching equation are better (one fit for

each data set). Thus, we also fit the generalized matching equation to each data set separately. The alternative hypothesis is shown graphically by the dashed and dotted lines in the left panel of Figure 10.11. The parameter values for our dataset without discriminative stimuli are $a = 0.80$ and $\log b = 0.10$; the parameter values for our dataset with discriminative stimuli are $a = 1.28$ and $\log b = 0.11$.

As always, before moving on to model comparison using an F test or the AIC, one must evaluate each model separately in terms of visual inspection, R^2 , residuals, and the other criteria listed earlier. It would not be appropriate to perform model comparison if one of the models produces nonrandom residuals or unrealistic parameter values (e.g., Dallery et al., 2005). Because no problems occurred with either the simpler or the alternative models (not shown), we can move on to statistical comparison. We calculate the residual sum-of-squares for the two-parameter (one fit with two parameters) global model and the more complicated four-parameter (two fits each with two parameters) alternative model. The total residual sum-of-squares for the simpler hypothesis, that a single global fit provides a better account of the data, is 0.71. The alternative hypothesis, that two separate fits of the equation provide a better account of the data, has a sum of squares of 0.34. The value of 0.34 was calculated by summing the individual fit sums-of-squares of 0.13 and 0.21. Next, we calculate the degrees of freedom for each model. The degrees of freedom for the global fit is 18 (20 data points minus two parameters); for the alternative hypothesis, it is 16 (20 data points minus four parameters—i.e., two parameters for each fit). As before, we can now calculate the relative difference in the sums-of-squares $([0.71 - 0.34]/0.34 = 1.09)$ and degrees of freedom $([18 - 16]/16 = 0.125)$ to allow the calculation of an F ratio. The F ratio is 8.72 $(1.09/0.125)$, with a numerator degree of freedom of 2 (the simpler hypothesis degree of freedom minus the alternative hypothesis degree of freedom) and a denominator degree of freedom of 16 (the alternative hypothesis degree of freedom), which has a p value of approximately .003. These calculations are summarized in Table 10.8. Assuming a threshold value of .05, we reject the simpler hypothesis and conclude that separate fits of the equation provide a better account of

TABLE 10.7

Log Response and Reinforcer Ratios Obtained Under Conditions With and Without Discriminative Stimuli

Without discriminative stimuli		With discriminative stimuli	
Log rnf ratio	Log rsp ratio	Log rnf ratio	Log rsp ratio
0.11	0.17	−0.3	−0.22
−0.37	0.07	−0.25	−0.12
0.49	0.61	1	1.6
−0.86	−0.66	−0.95	−0.99
0.7	0.63	0.43	0.62
−0.17	−0.21	−0.84	−1.2
0.63	0.57	−0.77	−0.83
−0.22	−0.06	0.43	0.43
0.25	0.24	−0.85	−0.85
−0.67	−0.44	0.06	0.08

Note. rnf = reinforcer; rsp = response.

TABLE 10.8

Extra Sum-of-Squares F Test Comparison of a Single Global Fit of the Generalized Matching Equation (Equation 5) With Both Data Sets in Table 10.7 Versus Two Separate Fits of the Generalized Matching Equation, One for Each Data Set

Model	SS	df
Simpler hypothesis (one global fit)	0.71	18.000
Alternative hypothesis (separate fit for each data set)	0.34	16.000
Difference	0.37	2.000
Relative difference	1.09	0.125
Ratio (F)	8.72	
p	.003	

the data than a single global fit. We can also see that the slope increased when discriminative stimuli were present and, as suggested by the difference in the intercepts, perhaps there were some small biases for certain colors over others.

A model comparison approach could also be used to assess whether a particular parameter value changed as a result of some intervention (for a more in-depth discussion of this technique, see Chapter G of Motulsky & Christopoulos, 2004). In the example just described using discriminative stimuli, we could have asked whether just the sensitivity parameter changed as opposed to the entire function. The general procedure is similar to what we have described in this section. Construct two models, a simpler and an alternative hypothesis, and ask whether the residuals are random, whether the parameters are sensible, and so on. In constructing the simpler model, again use a global fitting procedure, but with one key difference: Fit the equation to each dataset separately with the constraint that the value of the parameter of interest is shared between the two fits (e.g., the sensitivity parameter, a), and the other parameters are allowed to vary across the two fits. Least-squares regression in this case involves finding one value of the shared parameter and separate values of the other parameters, one for each data set, that minimize the total residual sum-of-squares. As before, the alternative model involves completely separate fits of the equation to

each dataset. Because a parameter value must be shared in the simpler model, this technique requires a program that can calculate least-squares regression with shared parameters, such as GraphPad Prism (GraphPad Software, Inc., La Jolla, CA) or Microsoft Excel. Finally, as with the previous model comparison example, compare the sum-of-squared residuals and the complexity (degrees of freedom) obtained for both models, calculate the F ratio, and evaluate significance using an F table. Following the same procedure outlined earlier, the AIC could also be used to evaluate whether an intervention changed the value of a parameter.

CONCLUSION

In this chapter, we introduced and developed the mechanics of how quantitative models are assessed and compared. This knowledge should be coupled with an appreciation of historical, conceptual, and philosophical issues concerning quantitative modeling in science in general and behavioral science in particular (e.g., Critchfield & Reed, 2009; Kline, 1959; Marr, 1989; Moore, 2008, 2010; Nevin, 1984; Rodgers, 2010; Smith, 1990). Space limitations preclude us from discussing these philosophical issues beyond acknowledging them, but we hope the reader is now better equipped to consider quantitative modeling from a conceptual and philosophical perspective and thereby bring quantitative modeling into the broader context of how science is conducted. Indeed, just as models of phenomena in other sciences have evolved on the basis of new empirical evidence, models in behavioral science have and will evolve with the accumulation of empirical evidence. We have emphasized that the precise description and critical evaluation of behavior's controlling variables are at the heart of quantitative analysis in behavioral science. Evaluating quantitative models requires clear and critical thinking and some specialized tools such as residual analysis and model comparison approaches. The benefits of quantitative models are considerable; they yield new insights into the causes of behavior and allow for a high degree of precision, falsifiability, and generality.

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